



Sustainability Capstone: Decarbonising Travel

Turning Scope 3 travel data into a quantified decarbonization roadmap.

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Executive Summary

114M kg

Total CO₂/yr from travel

Largest identifiable Scope 3 category.

24%

Emissions from exceptions

Almost a quarter of emissions stem from just 18% of trips booked out-of-policy.

90M

Travel Expenditure

Total financial footprint

~26%

Recoverable footprint

Recoverable annually via three quantified, actionable quick wins.

Top Insights

- 1. Premium Cabins:** Business/First Class are 25% of flights but contribute to the largest total emission (45%)
- 2. Out-of-Policy Trips:** 44% more carbon-emissive driven by late-bookings
- 3.** Asia, Oceania and South American regions are the biggest hotspots

Top Recommendations

- 1. Cabin Class:** Downsizing to Economy Class Flights on eligible routes can save roughly 4000 kg CO₂ per trip
- 2.** Electrify the rental car fleet to achieve an estimated 600 kg CO₂ reduction per trip
- 3. Prioritize Rail:** For short-haul routes Eg. Berlin-Madrid can save over 6500 kg CO₂ per trip and cut cash costs also

Why this matters: Analysis discovered certain factors that lie in our hands to control Scope 3 emissions

Emissions Profile & Benchmarking

- ▶ **Total Footprint:** 178M CO₂ generated across 65,289 trips (Dec 2023 – Jun 2025).
- ▶ **Industry Benchmark:** Our ~114,300t/yr run-rate aligns closely with the 140,000–150,000 t/yr baseline of comparable mid-size tech firms.
Source - Global Data
- ▶ **Mode & Class Intensity:** Business class emits roughly 3x more than economy, and first class emits 4x more each trip.
- ▶ **Ground Transport:** Rail and car travel remain a cleaner mode of commute than any flight class.



Hotspot Identification



Regional Benchmarking

Asia is the most carbon-intensive destination (~2,960 kg CO₂e/trip vs NA's ~1,540 kg) due to longer routes and limited rail options.



Absolute Global Hotspots

Oceania (89,400 t) and South America (39,700 t) account for >70% of the total global footprint, heavily driven by long-haul hubs.

Key Takeaway: No single culprit, footprints are broadly diversified across regions and units so, fixing the travel policy will largely outperform targeting any one team



Business Unit Drivers

Sales (60,100 t) and Marketing (33,400 t) generate 51% of total emissions, driven strictly by trip volume rather than inefficiency.



Travel Purpose

Customer Visits and Conferences combine for 79% of the footprint, presenting the largest target audience for behavioral change.

Inefficiencies & Pain Points

Process data shows the emission problem is linked to booking behaviour not approval bottleneck

- ▶ **Late Booking Root Cause:** Compliant trips average 22.1 days lead time vs. just 6.2 days for out-of-policy trips (forcing higher-carbon fares).
- ▶ **Dirty Out-of-Policy Travel:** Representing only 18.4% of trips, they generate 24.4% of emissions
- ▶ **Approval vs Booking:** Median approval time is just 3.4 hours. Automation should target the initial booking behavior, not the approval queue.
- ▶ **Disruption Penalties:** Thousands of mid-trip disruptions force last-minute rebooking, which rarely defaults to the cleaner option.

Strategic Recommendations



Right-Size Cabin Class

Shift eligible Business and First Class flights to Economy.

Impact: -24,900 tCO₂e/yr (Cost-neutral)



Electrify Rental Fleet

Replace diesel/petrol rentals with EVs across top operating markets.

Impact: -4,650 tCO₂e/yr (+\$430K/yr via leasing)



Rail-First

Mandate rail for high-frequency short routes like Berlin–Madrid.

Impact: -300+ tCO₂e/yr (Saves \$79K/yr)



Enforce 14-Day Window

Implement automated pre-approval blocks for flights booked under 14 days to directly mitigate policy leakage.

Impact: Targets 43,500 tCO₂e

Strategic Integration Roadmap

Phase 2

2027–28: Integrate & Automate

Embed live CO₂e estimates into the booking flow and automate the 14-day pre-approval gate to definitively close policy leakage.

Phase 1

2026–27: Quick Wins

Enforce cabin-class policies, pilot EV rentals in top 3 markets, and launch rail-first for top 10 EU routes (~30M CO₂e/yr reduction).

Phase 3

2029–30: Scale & Offset

Expand rail-first policy to Asia/NA assuming infrastructure permits by then, and utilize verified offsets for residual long-haul travel to achieve Net Zero.

Dashboard Tab 1

COORD: 34.9N / 12.4E SYSTEM INTERFACE // SOCIAL_OS_V50CB

UPTIME: 44:12:09

SOCIAL FEED

MESSAGES: 12

Dashboard



Overview

Insights

Process Analysis

Travel Sustainability Overview

Total Trips

65289

Total CO2 Emissions

178M kg

Total Cost

140M

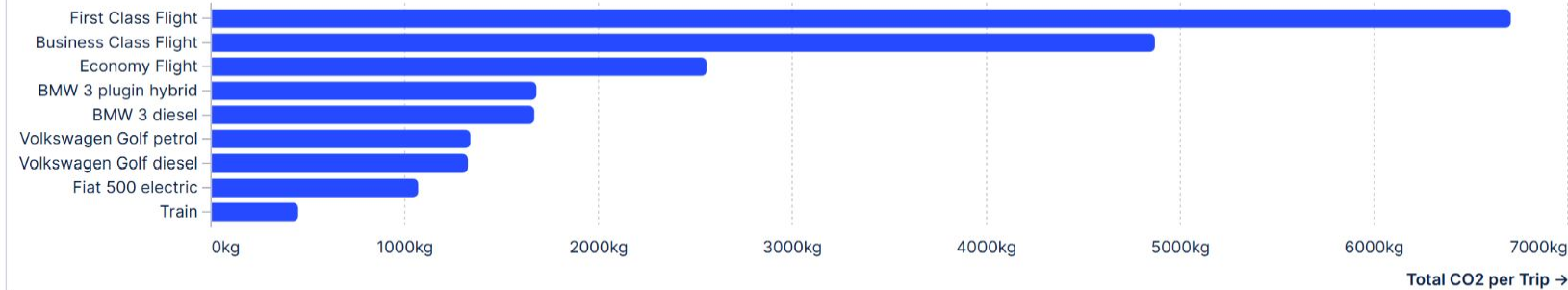
Avg CO2 per Trip

2,727.72 kg

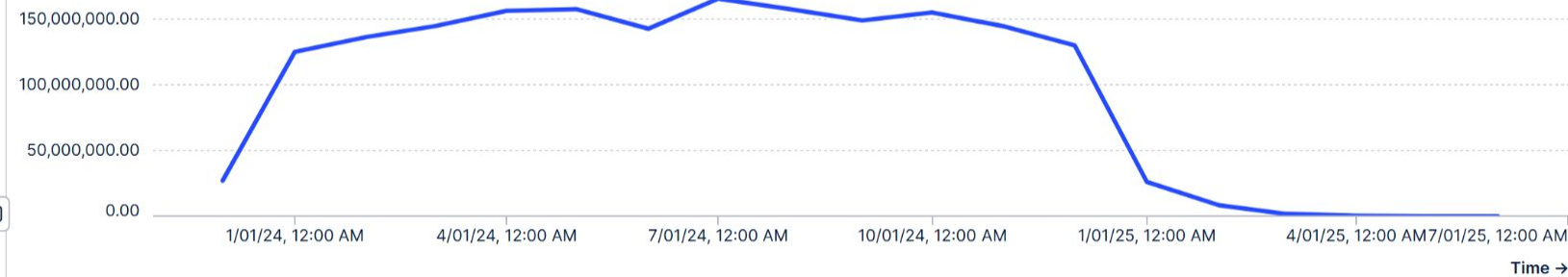
% Trips Out-of-Policy

18.39%

↑ Mode of Commute



↑ Total CO2



Filter bar

Route

Business Unit

Select Date

Dashboard Tab 2

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UPTIME: 44:12:09

SOCIAL FEED

MESSAGES: 12

Dashboard

Overview

Insights

Process Analysis

100%

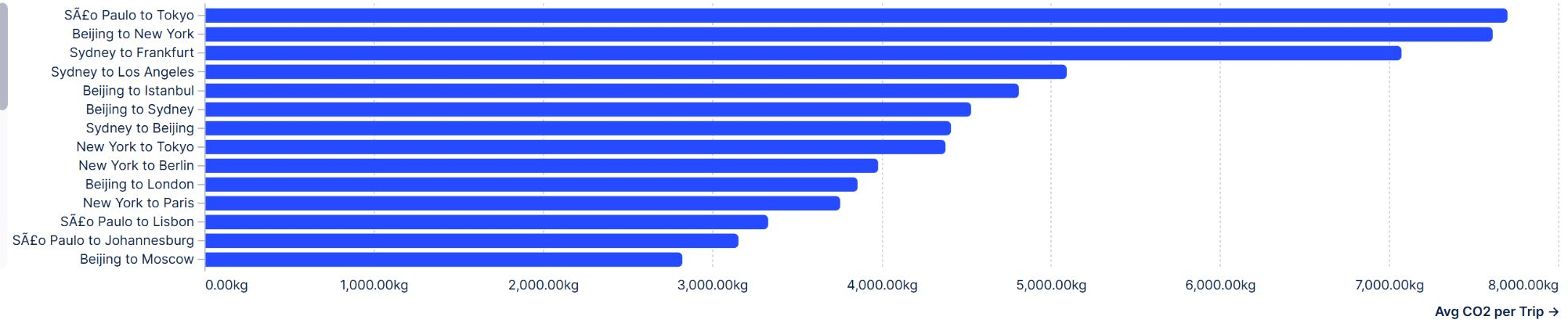
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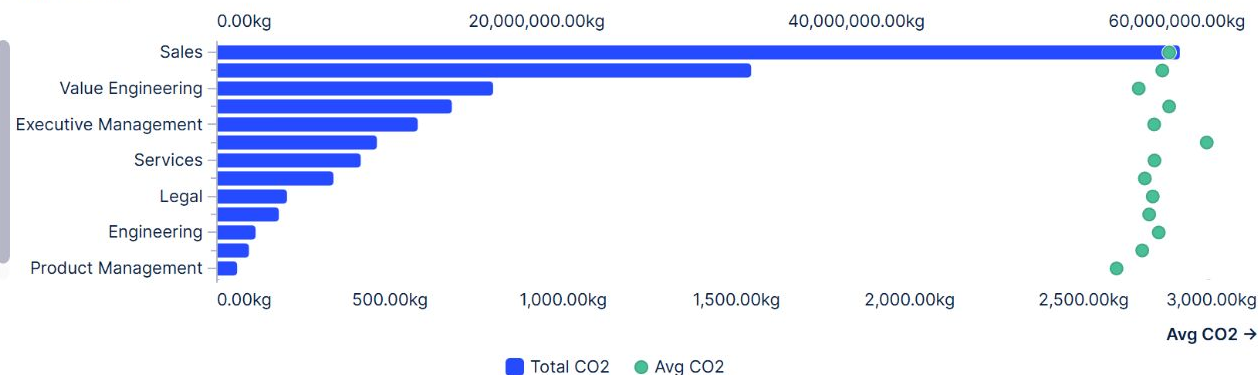
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Travel Sustainability Insights

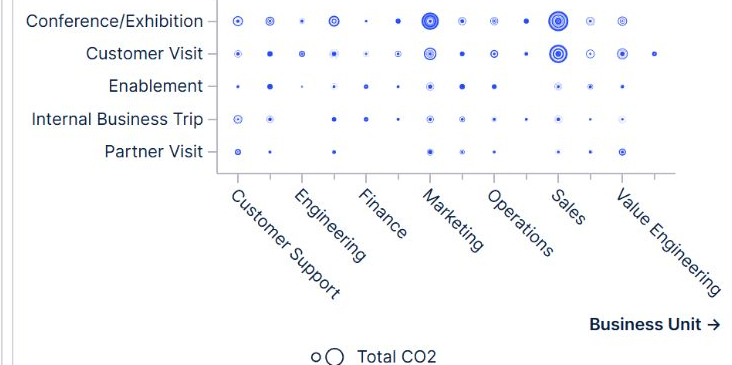
↑ Route



↑ Business Unit



↑ Purpose



Dashboard Tab 3

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Dashboard



Overview

Insights

Process Analysis



Process Analysis

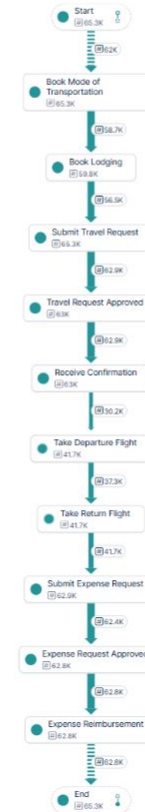
34 35% covered



Legend

43% +

34



Legend

41% +